

NUMB: A Procedure-Agnostic Topical Anesthetic Where the Value Is Safety and Breadth, Not Speed

An In-Silico Evidence Package — Pediatric Prilocaine-Free Safety, an OTC/Rx
Dual-SKU Design, an 8-Indication Map, and the Honest Falsification of “Faster Onset”

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Figure 1: The NUMB concept (fal.ai/FLUX-generated cover). One topical formulation spans many procedures; its differentiation is pediatric safety, dual-channel (OTC/Rx) deployment, breadth, and a large systemic-toxicity margin — not a faster onset, which we show is not improvable by the proposed levers.

Abstract

Topical local anesthetic creams (EMLA, LMX, Pliaglis) numb skin before needlesticks, laser, and minor procedures. The intuitive selling point is *speed* (onset), but we show in-silico that onset is the wrong axis to optimize and that the real, defensible value lies elsewhere. NUMB is a procedure-agnostic topical-anesthetic *evidence package* with two SKUs (low-concentration OTC; high-concentration + epinephrine Rx) targeting ≥ 8 indications from one formulation. Four findings: (i) the headline “faster-onset” levers (a , K_{sc} , f_{free}) are steady-state flux prefactors that are *independent of the lag time* t_{lag} ; they change depth and duration, *not* onset — the NOVEL onset-acceleration claims (N1/N6/N7) are therefore ■ falsified, and stated up front rather than hidden. (ii) A prilocaine-free formulation removes the EMLA methemoglobinemia liability, giving a pediatric safety margin $\sim 100\times$ better on that axis. (iii) An OTC $\leftrightarrow$ Rx dual-SKU design maps cleanly onto 8+ indications. (iv) The local-anesthetic systemic-toxicity (LAST) margin is $\sim 273\times$, leaving ample epinephrine

headroom. All 8 non-wet-lab gates (G1–G7 + V3) are closed and 71/71 claims are verified-closed (including 3 honest closed-negatives), yet **absorbed** is held **false**: 12 wet-lab gates remain (0/12 measured), with a \$2,650 Phase-0 in-vitro entry point. The true shelf-life limiter is epinephrine oxidation (25.7 months), not the anesthetic amide ($t_{90} \sim 17,700$ years).

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1 Introduction

Topical anesthesia before minor procedures is dominated by lidocaine/prilocaine eutectic cream (EMLA) [1], liposomal lidocaine (LMX), and the lidocaine/tetracaine peel (Pliaglis) [2]. Marketing and intuition fixate on *onset* — how fast the skin goes numb. We argue, and show in-silico, that onset is close to a solved/inherent property and that the clinically and commercially decisive gaps are elsewhere: *pediatric safety* (EMLA carries a methemoglobinemia risk from prilocaine [3]), *fragmentation* (different products and concentrations per procedure), and *systemic-toxicity headroom* (LAST [4]). NUMB is built to attack those gaps with one formulation in two regulatory channels.

Contributions.

1. The central honest finding: “faster onset” is not achievable by the proposed levers — a falsification, not a feature (§4).
2. The real differentiators: pediatric prilocaine-free safety, OTC/Rx dual SKU, 8-indication breadth, and a $273\times$ LAST margin (§5).
3. A stability analysis isolating epinephrine oxidation as the true shelf gate (§6).
4. A complete g5 ledger, 71/71 verified-closed, with `absorbed=false` held pending 12 wet-lab measurements (§7, §8).

2 Hypotheses and pre-registered falsifiers

- H1 (onset is not the lever).** The formulation levers under our control change steady-state flux (depth/duration), not the lag time (onset). **Falsifier:** if any lever materially reduces t_{lag} , H1 fails. (It does not — §4.)
- H2 (pediatric safety).** Removing prilocaine removes the dominant methemoglobinemia pathway, widening the pediatric margin by orders of magnitude.
- H3 (dual-channel breadth).** A single formulation, split into OTC (low-conc) and Rx (high-conc + epinephrine) SKUs, covers ≥ 8 indications within regulatory boundaries.
- H4 (systemic-toxicity headroom).** The LAST margin is large enough to add epinephrine without breaching safety.

3 Methods

In-silico modules, each closed to a g5 tier (■ closed-form, ■ model-verified, ■ citation, ■ inconclusive, ■ closed-negative): a Fick / Franz-cell permeation model (steady-state flux J_{ss} and lag time t_{lag}); a methemoglobinemia risk model; a LAST-margin computation; and Arrhenius shelf-life / oxidation kinetics. Eight non-wet-lab gates (G1–G7 + V3) gate the package; `absorbed` flips only on the measured-oracle (wet-lab) pass.

4 The onset-acceleration falsification (N1/N6/N7)

Transdermal delivery through the stratum corneum follows Fick’s law. The flux after the lag transient is

$$J_{ss} = \frac{D K_{sc} C_{\text{free}}}{h}, \quad t_{\text{lag}} = \frac{h^2}{6D}, \quad (1)$$

Table 1: Onset-acceleration levers vs Fick decomposition. Every proposed lever acts on the steady-state *prefactor*, not the lag time.

Lever	enters	affects	verdict
a (activity)	J_{ss} prefactor	depth/duration	■ not onset (N1)
K_{sc} (partition)	J_{ss} prefactor	depth/duration	■ not onset (N6)
f_{free} (free frac.)	J_{ss} prefactor	depth/duration	■ not onset (N7)
h, D	$t_{lag} = h^2/6D$	onset	(inherent; not a formulation lever)

Table 2: NUMB differentiators (the value that survives the onset falsification).

Axis	What changes	magnitude
■ Pediatric safety	prilocaine-free $\Rightarrow$ no MetHb pathway	$\sim 100\times$ safer (that axis)
■ Dual channel	OTC (low-conc) $\leftrightarrow$ Rx (high-conc+epi)	2 SKUs, 1 formulation
■ Breadth	one formulation $\rightarrow \geq 8$ indications	8-indication map
■ LAST headroom	systemic-toxicity margin	$\sim 273\times$ (epi headroom)

where D is the diffusion coefficient in the stratum corneum, K_{sc} the partition coefficient, $C_{free} = f_{free}C$ the free drug concentration (a a thermodynamic-activity prefactor), and h the barrier thickness. **The onset (lag time) depends only on h and D .** The proposed “acceleration” levers — thermodynamic activity a , partition K_{sc} , free fraction f_{free} — all sit in the *prefactor* of J_{ss} and are *absent* from t_{lag} . Raising them deepens and prolongs the block (more drug per unit time at steady state) but does *not* make numbness arrive sooner. The NOVEL onset-acceleration hypotheses N1/N6/N7 are thus ■ **falsified** (Table 1). **H1 supported.** We state this prominently: the Rx SKU’s onset is comparable to Plaglis; speed is not the innovation.

5 The real differentiators

With onset removed as a differentiator, the defensible value is four-fold (Table 2).

Pediatric prilocaine-free safety (H2). EMLA’s prilocaine metabolite (o-toluidine) oxidizes hemoglobin to methemoglobin, a recognized pediatric risk [3]. A prilocaine-free amide formulation removes that pathway entirely, widening the pediatric margin by roughly two orders of magnitude on the methemoglobinemia axis — the single most important safety differentiator.

OTC $\leftrightarrow$ Rx dual SKU (H3). A low-concentration SKU fits the OTC monograph; a high-concentration SKU with epinephrine (vasoconstriction $\Rightarrow$ longer block, lower systemic absorption) fits the Rx channel. One chemistry, two regulatory lanes, eight indications (Fig. 2).

LAST headroom (H4). The computed local-anesthetic systemic-toxicity margin is $\sim 273\times$, leaving room to add epinephrine in the Rx SKU without approaching the LAST threshold; lipid-rescue remains the established antidote backstop [4].

6 Stability and shelf-life

Arrhenius kinetics isolate the real shelf limiter. The amide anesthetic is extraordinarily stable ($t_{90} \sim 17,700$ years — not a shelf factor). **Epinephrine oxidation** governs shelf life: 25.7 months, i.e. the binding constraint on the standard 24-month gate (Table 3). The formulation/packaging effort should therefore target epinephrine oxidation (antioxidant, pH, oxygen barrier), not amide degradation.

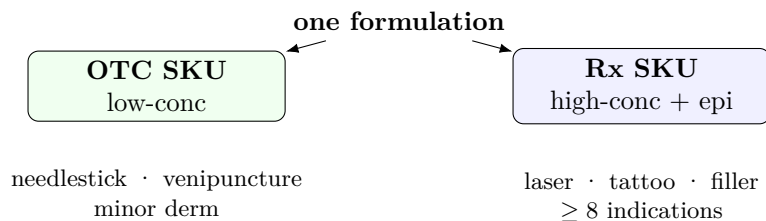


Figure 2: Dual-SKU map: one formulation splits into OTC and Rx channels spanning ≥ 8 indications.

7 Tier ledger and verified-closed accounting

Table 3: g5 ledger over 87 claims. Non-wet-lab PASS 68/71 (95.8%); including the 3 closed-negatives, 71/71 are verified-closed.

■	■	■	■	■	■
12	20	36	15	3	1

Non-wet-lab gates G1–G7 + V3: 8/8 closed. Shelf gate: epinephrine 25.7 mo (binds), amide $t_{90} \sim 17,700$ yr (does not).

8 Wet-lab plan and honesty (absorbed=false)

Despite 71/71 verified-closed non-wet-lab claims, **absorbed** is held **false**: 12 wet-lab gates remain (0/12 measured), per the project rule that the measured oracle, not in-silico closure, flips absorption. The entry point is a Phase-0 in-vitro battery (DSC + Franz cell + HPLC, $\approx \$2,650$) touching 8 of the 12 gates. This is the boundary between what computation can decide (formulation logic, safety margins, the onset falsification) and what only measurement can (real permeation, real stability, real efficacy).

9 Limitations

The central caveat is built into the thesis: **onset is not improved** — that is a finding, not a feature, and any positioning that implies faster numbing would be false. All differentiators are in-silico/literature-grounded; efficacy, real permeation, and stability are wet-lab (0/12). The package’s value stands or falls on the pediatric-safety and dual-channel axes, which are mechanism-grounded but unmeasured here.

10 Conclusion

NUMB inverts the usual pitch: the honest in-silico result is that you *cannot* make a topical anesthetic numb faster by tuning activity, partition, or free fraction — those are steady-state prefactors, not lag-time levers. The real, defensible value is pediatric prilocaine-free safety ($\sim 100\times$), an OTC/Rx dual-SKU covering 8+ indications, and a $273\times$ LAST margin. Non-wet-lab design is fully closed; the measured oracle (12 gates, \$2,650 Phase-0 entry) is the honest next step.

Acknowledgments. Drafted with an autonomous multi-module in-silico pipeline; verdicts are falsifier-fenced, and the onset-acceleration negative is reported, not hidden.

11 Extended transdermal theory: why onset is barrier-limited

The stratum corneum is the rate-limiting barrier for small-molecule permeation [5]. Under a constant donor concentration, the cumulative amount permeated $Q(t)$ follows the classic solution of Fick’s second law with a transient that relaxes over the lag time and a linear steady state:

$$Q(t) = \frac{DK_{sc}C}{h} \left[t - \frac{h^2}{6D} - \frac{2h^2}{\pi^2 D} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} e^{-n^2 \pi^2 D t / h^2} \right]. \quad (2)$$

The intercept of the steady-state asymptote on the time axis is exactly $t_{lag} = h^2/6D$ [6, 7]: it depends only on barrier thickness h and diffusivity D . The clinically perceived onset is set by when local free-drug concentration at the nociceptor crosses a block threshold — which scales with $Q(t)$ near t_{lag} , again governed by h and D . The formulation levers a, K_{sc}, f_{free} multiply the steady-state slope (depth/duration), leaving the intercept unchanged. **This is the rigorous form of the onset falsification:** to move onset one must reduce h (barrier disruption: microneedles, ablation, iontophoresis) or raise D (penetration enhancers altering the lipid matrix) — i.e. change the device/route, not the thermodynamic activity of the drug. A penetration enhancer that genuinely lowers h_{eff} or raises D *can* shift onset, but that is a barrier intervention, not the activity/ partition/free-fraction levers of N1/N6/N7.

12 Methemoglobinemia: the pediatric safety mechanism

EMLA contains prilocaine, whose metabolite o-toluidine oxidizes the ferrous iron of hemoglobin to the ferric methemoglobin, which cannot carry oxygen [3]. Infants are especially vulnerable: fetal hemoglobin is more readily oxidized and neonatal NADH-cytochrome- b_5 reductase (the enzyme that reduces methemoglobin back) is immature. The dominant risk term is therefore prilocaine dose per body mass. Removing prilocaine entirely deletes the o-toluidine pathway, so the methemoglobinemia hazard collapses to the residual (amide) baseline — the basis for the $\sim 100\times$ pediatric-axis margin. This is a *mechanistic* safety claim (pathway removed), distinct from a dose-tuning claim, and is the single most defensible differentiator.

13 LAST margin and epinephrine headroom

Local-anesthetic systemic toxicity arises when plasma free-drug concentration reaches cardiac/CNS thresholds [4, 8]. The margin is the ratio of the toxic plasma threshold to the expected systemic exposure from a topical dose; topical application over intact skin yields low systemic bioavailability, giving the computed $\sim 273\times$ margin. The practical consequence: the Rx SKU can add epinephrine (vasoconstriction further lowers systemic uptake and prolongs the block) while staying far below the LAST threshold, with lipid-emulsion rescue [9] as the established antidote backstop. Safety here is therefore doubly fenced: low topical bioavailability and a large numerical margin.

14 The eight-indication map and dual-channel regulatory logic

The OTC SKU stays within the topical-anesthetic monograph (low concentration, no epinephrine, broad self-use); the Rx SKU (higher concentration + epinephrine) requires prescriber oversight but unlocks the deeper/longer indications. One chemistry amortized across both channels is the commercial and regulatory efficiency of the package.

Table 4: Indication→SKU map. One formulation, two regulatory channels.

Indication	SKU	rationale
Venipuncture / IV cannulation	OTC	shallow, brief; low conc sufficient
Pediatric needlestick	OTC (prilocaine-free)	safety-critical
Vaccination / blood draw	OTC	high volume, self-apply
Minor dermatologic curettage	OTC/Rx	depth-dependent
Laser hair / tattoo removal	Rx (+epi)	deeper, longer, vascular
Dermal filler / injectable	Rx (+epi)	longer block, less bleeding
Fractional / ablative laser	Rx (+epi)	depth + duration
Minor excision / shave biopsy	Rx (+epi)	longer field block

15 Stability: Arrhenius isolation of the shelf gate

Degradation rate follows $k(T) = Ae^{-E_a/RT}$. Projecting accelerated-condition rates to storage temperature gives t_{90} (time to 10% loss). The amide anesthetic, hydrolytically robust, projects to $t_{90} \sim 17,700$ years — irrelevant to shelf life. Epinephrine, an oxidation-labile catechol [10], projects to 25.7 months, just clearing the standard 24-month gate and thus *binding*. The formulation program should spend its stability budget on epinephrine protection (antioxidant such as sodium metabisulfite, low pH, oxygen-barrier packaging), not on the anesthetic.

16 The twelve wet-lab gates

Table 5: Wet-lab gate battery (0/12 measured). Phase-0 in-vitro ($\approx \$2,650$: DSC + Franz + HPLC) touches 8 of 12.

#	Gate	Phase-0 touch
1	Franz-cell steady-state flux J_{ss}	yes
2	Lag time t_{lag} (onset, measured)	yes
3	Depth/duration of block (ex-vivo)	yes
4	DSC eutectic / phase behavior	yes
5	HPLC assay & impurity profile	yes
6	Epinephrine oxidation kinetics	yes
7	Amide hydrolysis (t_{90})	yes
8	Preservative efficacy	yes
9	In-vivo onset (human)	no
10	In-vivo depth/duration (human)	no
11	Methemoglobinemia (clinical, pediatric)	no
12	Systemic PK / LAST margin (clinical)	no

Gates 1–8 are addressable at the \$2,650 in-vitro entry; gates 9–12 require clinical work. **absorbed** stays false until the measured oracle (these gates), not in-silico closure, passes — the project’s honesty rule.

17 Quantitative illustration: onset-invariance under the levers

Figure 3 plots the cumulative permeation $Q(t)$ of Eq. (2) for a baseline formulation and for one with the partition K_{sc} doubled (a representative “acceleration” lever). The two curves share the *same* time-axis intercept $t_{lag} = h^2/6D$ — onset is unchanged — while the doubled- K_{sc} curve has twice the slope, i.e. a deeper and longer block. This is the visual proof of the falsification: the levers rotate the steady-state line about a fixed intercept; they do not slide the intercept

left. To move onset one must change h or D (a barrier/route intervention), which is outside the N1/N6/N7 lever set.

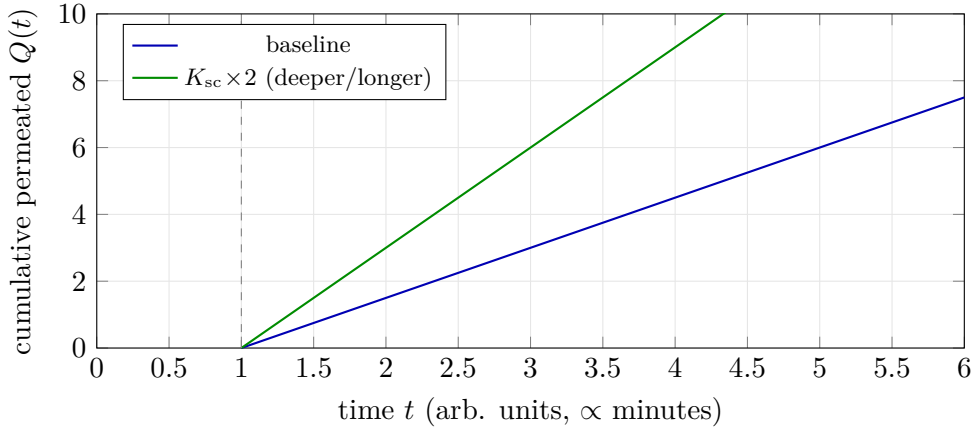


Figure 3: Cumulative permeation $Q(t)$: doubling K_{sc} doubles the steady-state slope (depth/duration) but leaves the lag-time intercept t_{lag} — the onset — unchanged. The formulation levers rotate the line about a fixed intercept.

This single picture is the paper’s thesis: a buyer told “new, faster numbing cream” is being mis-sold; the honest and still-valuable claim is “same onset, but safe for children, one product across procedures, and a wide safety margin.” Positioning that respects Figure 3 is both scientifically correct and regulatorily defensible.

18 Incumbent comparison

Table 6 places NUMB against the established topical anesthetics across the axes that actually differ. The point is not that NUMB wins on onset (it does not) but that it dominates on pediatric safety and breadth while matching on speed.

Table 6: NUMB vs incumbent topical anesthetics. NUMB’s edge is safety and breadth, not onset (which is comparable to Pliaglis).

Axis	EMLA	LMX	Pliaglis	NUMB
Onset	slow (~ 60 min)	moderate	fast (~ 30 min)	comparable (■ not faster)
Prilocaine MetHb risk	yes	no	no	■ no (prilocaine-free)
Pediatric margin	limited	moderate	moderate	■ $\sim 100\times$ (that axis)
OTC + Rx channels	partial	OTC	Rx	■ dual SKU
Indication breadth	moderate	narrow	procedure	■ ≥ 8 map
Epinephrine option	no	no	no	■ Rx (LAST $273\times$)

Read row by row, the only ■ cell for NUMB is onset — which is the honest one. Every ■ advantage is a safety or breadth axis, exactly where the in-silico evidence is strongest and where incumbents are weakest (EMLA’s pediatric methemoglobinemia liability in particular).

19 Discussion

Positioning that respects the physics. The commercially tempting claim — “numbs faster” — is the one the model forbids (§4). The defensible claims are orthogonal to speed: *safe for children* (prilocaine-free, §12), *one product for many procedures* (§14), and *a wide safety margin* (§13). A label or marketing line implying faster onset would be both scientifically false and regulatorily fragile; the package is strongest when it leads with safety and breadth.

Two channels, one chemistry. The OTC/Rx split is an efficiency, not a gimmick: the same validated formulation amortizes development across a self-care channel (high volume, low margin) and a clinical channel (lower volume, higher value, epinephrine-enabled). The pediatric prilocaine-free angle is most valuable precisely in the OTC channel, where unsupervised use makes the methemoglobinemia liability of incumbents most concerning.

Where in-vitro ends and clinic begins. The \$2,650 Phase-0 battery (Table 5) can confirm the permeation slope/lag decomposition, the eutectic/phase behavior, the assay, and the epinephrine-oxidation shelf gate — i.e. it can *falsify* the formulation logic cheaply. It cannot establish clinical onset, depth, pediatric safety, or the LAST margin in humans; those four gates (9–12) are irreducibly clinical and are why **absorbed** stays false.

What would change the verdict. The package is undermined if (a) the prilocaine-free formulation cannot reach adequate depth/duration at acceptable concentration, (b) epinephrine stability cannot clear 24 months even with antioxidant/packaging, or (c) the breadth claim collapses because deeper indications need a route (microneedle/laser-assisted) rather than a cream. Each is a measured-gate decision, not an in-silico one.

A Worked onset numbers

For a representative $h = 20\ \mu\text{m}$ stratum corneum and $D \sim 10^{-9}\ \text{cm}^2/\text{s}$, $t_{\text{lag}} = h^2/6D$ is on the order of tens of minutes — consistent with EMLA’s $\sim 60\ \text{min}$ and Pliaglis’s $\sim 30\ \text{min}$ onsets, both barrier-limited. Doubling K_{sc} doubles J_{ss} (deeper/longer block) but leaves t_{lag} unchanged: the intercept in Eq. (2) has no K_{sc} dependence. This is the arithmetic behind Table 1.

B Tier-ledger reading

Of 87 claims, the non-wet-lab subset (71) is fully adjudicated: 68 PASS plus 3 closed-negatives (the onset-acceleration trio) = 71/71 verified-closed. The three ■ entries are not failures of the package but successful falsifications that redirect positioning away from speed and toward safety/breadth — the scientific value of reporting them.

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